

## Module 8: Cellular Genetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. What is the main product of mitosis?
2. A. Four haploid gametes
3. B. Two genetically identical daughter cells
4. C. A single haploid cell
5. D. Cells with randomly scrambled DNA
6. Answer: B
  
7. Why: Mitosis produces two genetically identical daughter cells for growth and repair.  
Meiosis makes four haploid gametes.
  
8. Meiosis reduces the chromosome number by using:
9. A. two rounds of DNA copying and no division
10. B. no DNA replication at all
11. C. extra chromosomes added each cycle
12. D. one round of DNA copying followed by two divisions
13. Answer: D
  
14. Why: One replication followed by two divisions halves the chromosome number,  
producing haploid gametes.
  
15. Crossing over and independent assortment both act to:
16. A. increase genetic variation among gametes
17. B. produce identical gametes every time
18. C. occur only during mitosis
19. D. remove all differences between offspring
20. Answer: A
  
21. Why: Both processes shuffle alleles during meiosis, generating genetic variation among  
offspring.

22. If chromosomes fail to separate correctly during meiosis (nondisjunction), a resulting gamete may have:

23. A. exactly the normal number every time

24. B. no DNA whatsoever

25. C. an extra or a missing chromosome

26. D. twice as many cells

27. Answer: C

28. Why: Nondisjunction gives a gamete an abnormal chromosome count, which can lead to conditions such as trisomy.